

Furthermore, things can get even more surprising when randomness changes in shape, as with regime switches. A regime switch corresponds to situations when all of the attributes of a system change to the point of it becoming unrecognizable to the observer. Darwinian fitness applies to species developing over a very long time, not observed over a short term – time aggregation eliminates much of the effects of randomness; things (I read *noise*) balance out over the long run, as people say.

Owing to the abrupt rare events, we do not live in a world where things “converge” continuously towards betterment. Nor do things in life move *continuously* at all. The belief in continuity was ingrained in our scientific culture until the early twentieth century. It was said that *nature does not make jumps*; people quote this in well-sounding Latin: *natura no facit saltus*. It is generally attributed to the eighteenth-century botanist Linnaeus who obviously got it all wrong. It was also used by Leibniz as a justification of calculus, as he believed that things are continuous no matter the resolution at which we look at them. Like many well-sounding “make sense” types of statements (such dynamics made perfect intellectual sense), it turned out to be entirely wrong, as it was denied by quantum mechanics. We discovered that, in the very small, particles jump (discretely) between states; they do not slide between them.

CAN EVOLUTION BE FOOLED BY RANDOMNESS?

We end this chapter with the following thought. Recall that someone with only casual knowledge about the problems of randomness would believe that an animal is at the maximum fitness for the conditions of his time. This is not what evolution means; *on average* animals will be fit, but not every single one of them, and not at all times. Just as an animal could have survived because its sample path was lucky, the “best” operators in a given business can come from a subset of operators who survived because of over-fitness to a sample path – a sample path that was free of the evolutionary rare event. One vicious attribute is that the longer these animals can go without encountering the rare event, the more vulnerable they will be to it. We said that should one extend time to infinity, then, by *ergodicity*, that event will happen with certainty –

the species will be wiped out! For evolution means fitness to one and only time series, not the average of all the possible environments.

By some viciousness of the structure of randomness, a profitable person like John, someone who is a pure loser in the long run and correspondingly unfit for survival, presents a high degree of eligibility in the short run and has the propensity to multiply his genes. Recall the hormonal effect on posture and its signaling effect to other potential mates. His success (or pseudo-success owing to its fragility) will show in his features as a beacon. An innocent potential mate will be fooled into thinking (unconditionally) that he has a superior genetic makeup, until the following rare event. Solon seems to have gotten the point; but try to explain the problem to a naïve business Darwinist – or your rich neighbor across the street.